

SunoikisisDC Digital Approaches to Cultural Heritage
Spring 2026. Session 8

Ethics, Sustainability & Paradata in 3D Heritage

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Agenda

- Theoretical introduction: Ethics, Sustainability & Paradata and FAIR/CARE in 3D
- Sustainability (3 meanings)
 - Case study 1: Inventory of 3D methods in Dutch archaeology (Alicia)
 - Case study 2: H3DAR project / using commercial platforms for FAIR purposes. (Tom)
 - Discussion
- Ethics
 - Case study 1: practical vs. philosophical ethics (Alicia)
 - Case study 2: tDAR discussion / visual prototyping (Tom)
 - Discussion
- Paradata
 - Case study 1: H3DAR survey / interview responses & IIIF 3D (Tom)
 - Discussion

FAIR and CARE for 3D data management



Definitions

Sustainability	Long-term and future-proof management of data; environmental impact of technology use.
Paradata	Documentation on the process of creating or collecting data.
Ethics	Moral principles that govern our behaviour or actions.
<i>Pragmatic ethics</i>	Self-reflexive, contextual and historical.

Sustainability

Long term access & availability

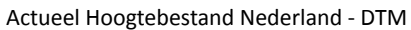
Case study 1

Inventory of 3D and remote sensing technologies in
Dutch archaeology: who, what, and why

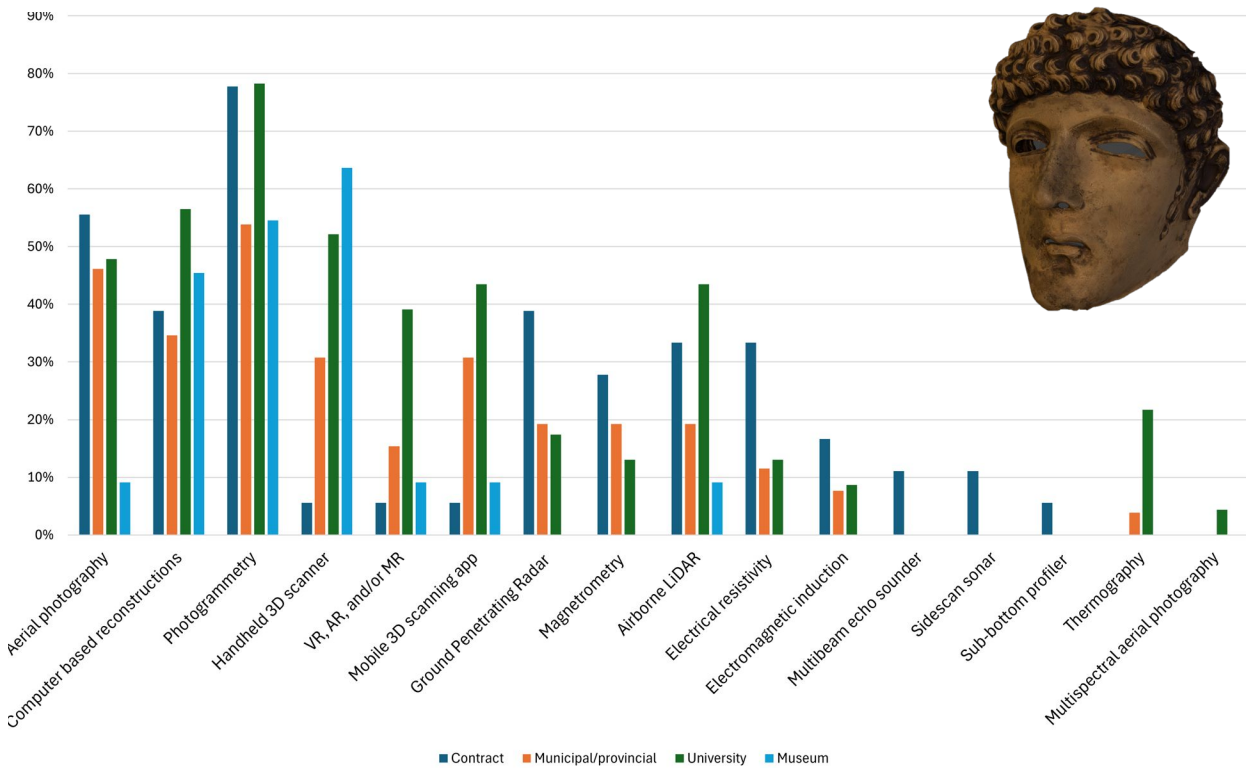
(Alicia Walsh, PhD Research)



Universiteit
Leiden

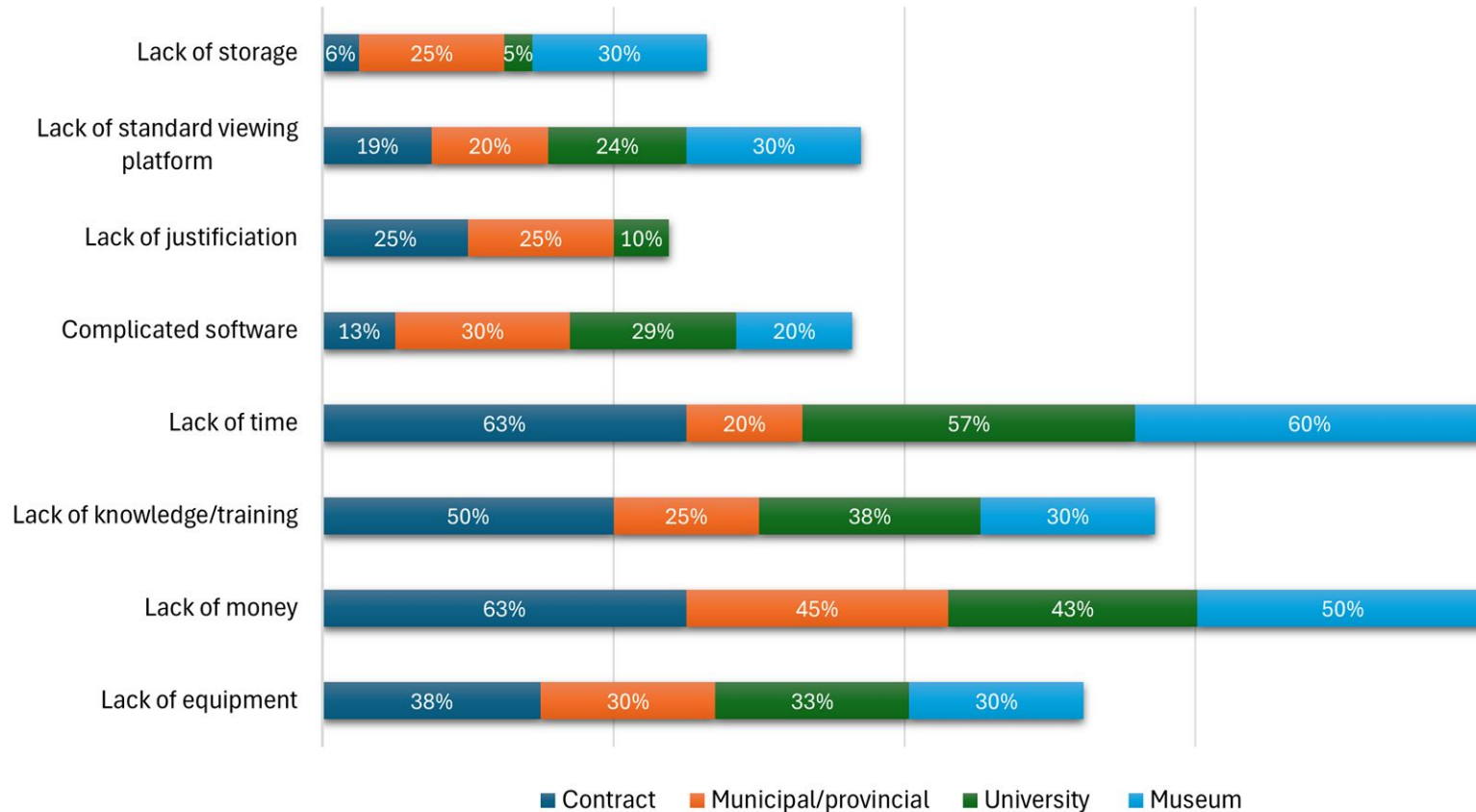


Which methods are being used?



Leiden archaeological field school - Oss. Photo: Arjan Louwen
3D image of Matilo mask- Erfgoed Leiden en Omstreken

Main challenges faced using digital technologies



FAIR 3D data

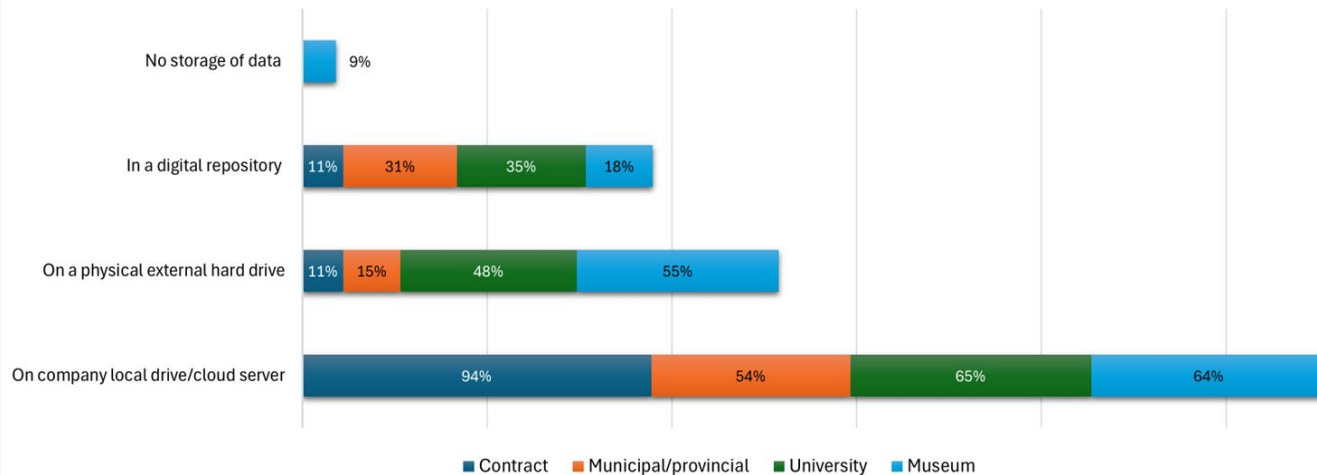
Is it findable and accessible?

Data Archiving and Networked Services



Sketchfab

Where digital data is stored



Accessibility of Digital Data

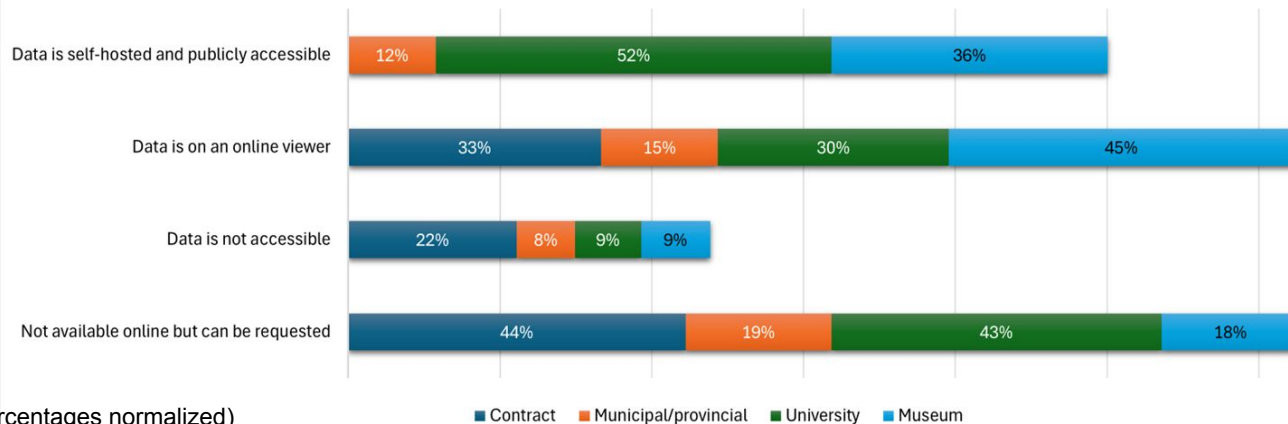
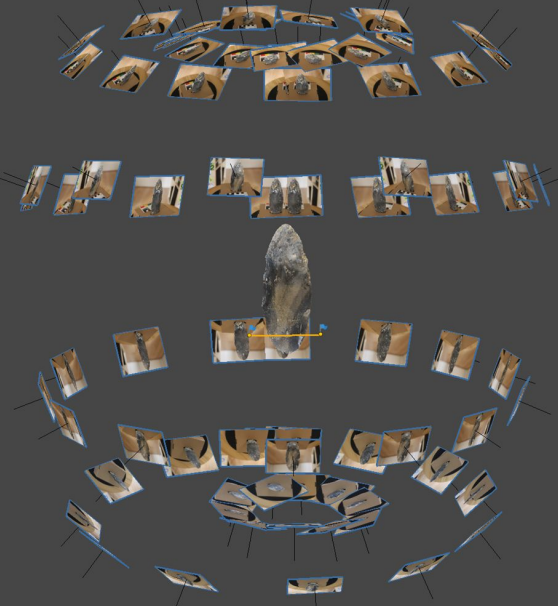


Image of 3D trench profile – Municipality of the Hague



Flint axe - 3D model by A. Walsh



What do we
save?

Sustainability

Long term access & availability

Case study 2

The UK Heritage 3D Data at Risk project

Benefits/risks of using commercial platforms
in pursuit of achieving FAIR goals.

Sustainability (Long term access & availability)

c.2010s

3D digitisation is becoming more accessible, useable, affordable. Heritage organisations are looking for ways to engage audiences with their collections.

2011

sketchfab.com launches in as the 'youtube for 3D'.

2015

Sketchfab begins to offer **free** + **discounted** accounts to cultural heritage organisations.

2021

Sketchfab [acquired by Epic Games](#).

2023

Sketchfab/Epic [announce](#) planned changes to services; concern in heritage sector and artist circles ensues.

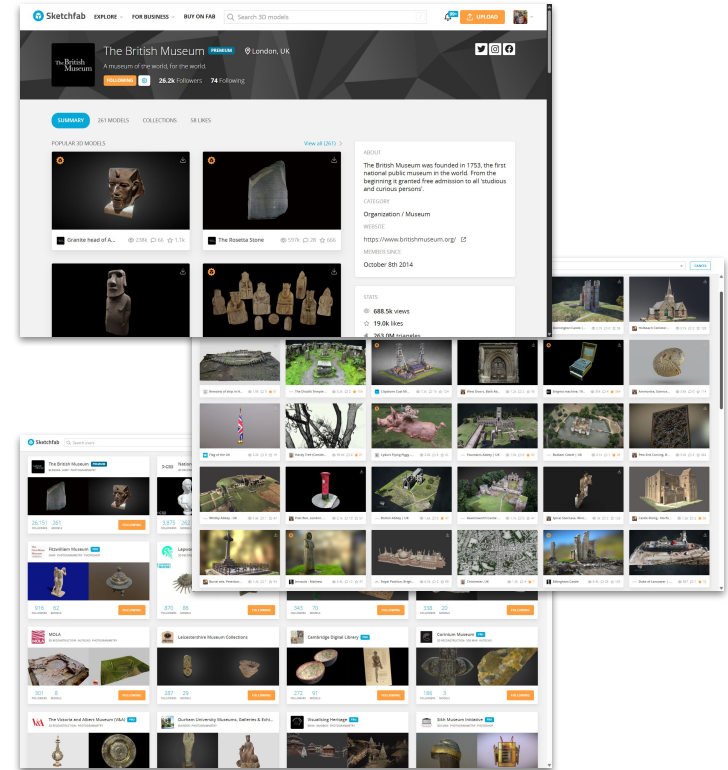
2025

[UK 3D Heritage Data at Risk](#) launched.

What's the problem?

Many UK heritage organisations rely on sketchfab.com for **publishing** and **sharing** 3D data online, as well as various **platform tools** (3D scene set up, annotations, AR & VR, etc.) for everything from **storytelling** to **scientific communication**.

While Sketchfab *is* a great product, **changes** being made its owners, a **lack of alternatives**, and **variable data governance practice** are causes for concern...



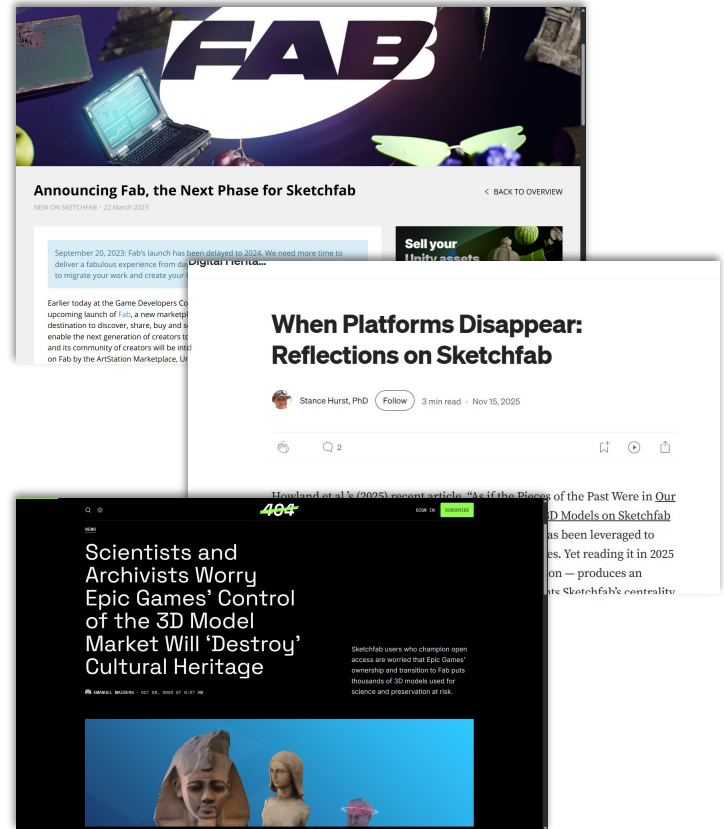
- **230+** UK heritage orgs
- **11,000+** 3D models
- **63,000+** followers

[source](#)

What's the project?

The aim of the project is to help safeguard access to the UK's digital 3D heritage at risk through over-reliance on commercial and free-to-use platforms for storage and dissemination.

The problem also isn't Sketchfab specific: using any 3rd party or commercial platform can carry similar risks.



How does Sketchfab help achieve FAIR data?

Findable

- SEO for user profiles, 3D model page, etc.
- Search, tagging, category features.
- Social features (follow, like, share, comment).

Accessible

- Data optimisation for web delivery.
- Web standards based 3D viewer (webGL).
- Desktop & mobile UI / UX.

Interoperable

- Data converted to modern standards e.g. glTF (GLB), USD.
- Data and 3D viewer access and customisation via application programming interfaces (APIs)

Re-usable

- Embeddable 3D viewer.
- Download feature.
- Open, free, & paid license framework.

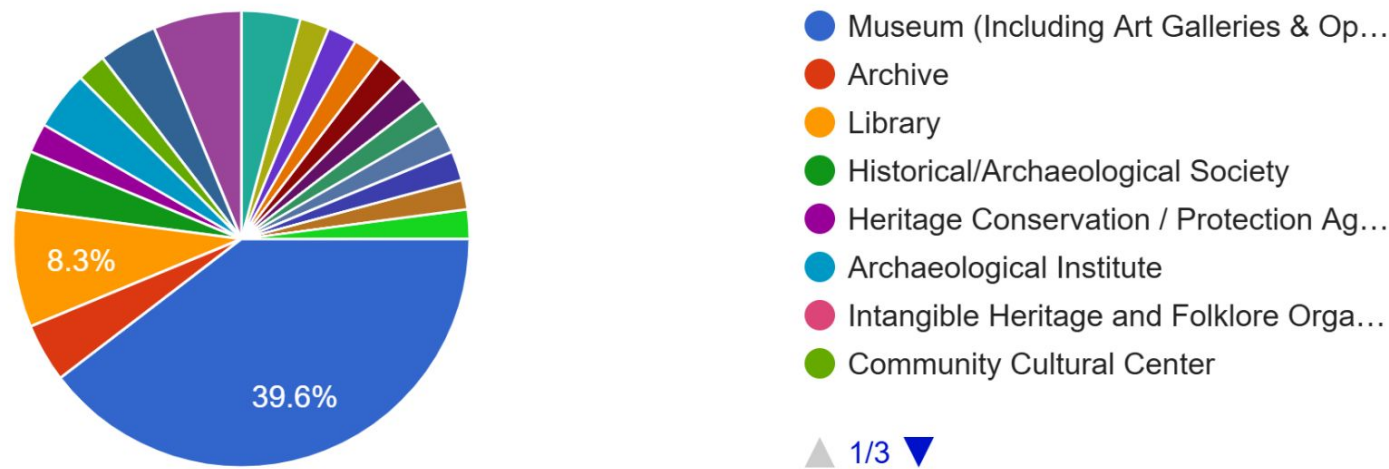
Free™, easy to use, access to new audiences, publish alongside peers, &tc...

User Research

- How and why are UK heritage organisations creating 3D data?
- What UK heritage sector needs does Sketchfab meet/not meet?
- What heritage specific alternatives to Sketchfab are available today?
- ***Is there a need for new UK heritage focussed 3D data services or platforms?***
- 230+ UK heritage sketchfab users identified, 11,000+ 3D models, 63,000+ followers
- Outreach to 140+
- To date: 48 survey respondents, 26 interviews, 2 online events.

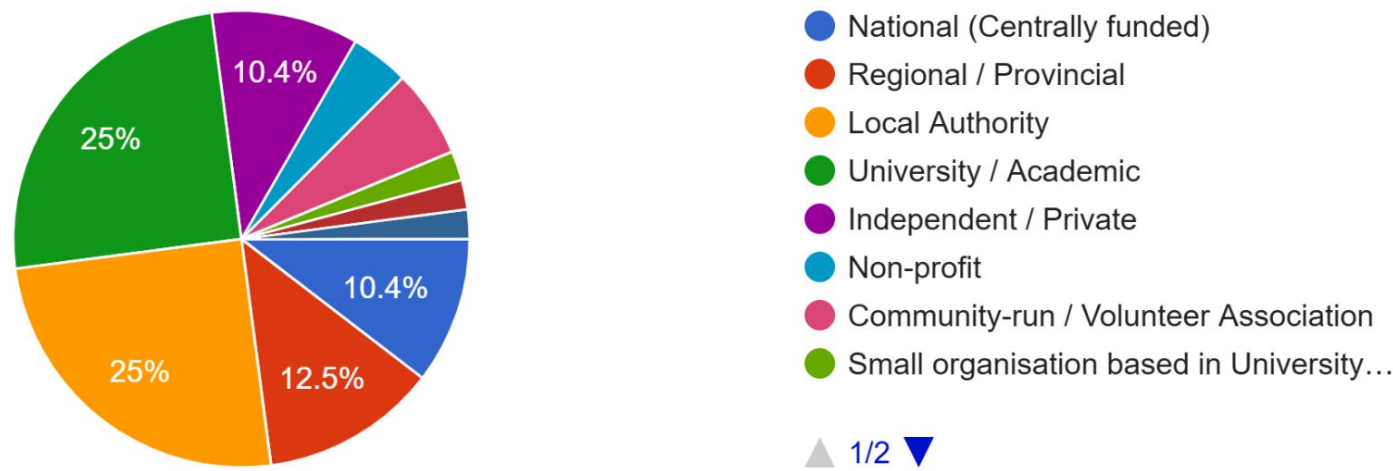
What type of organisation do you represent?

48 responses



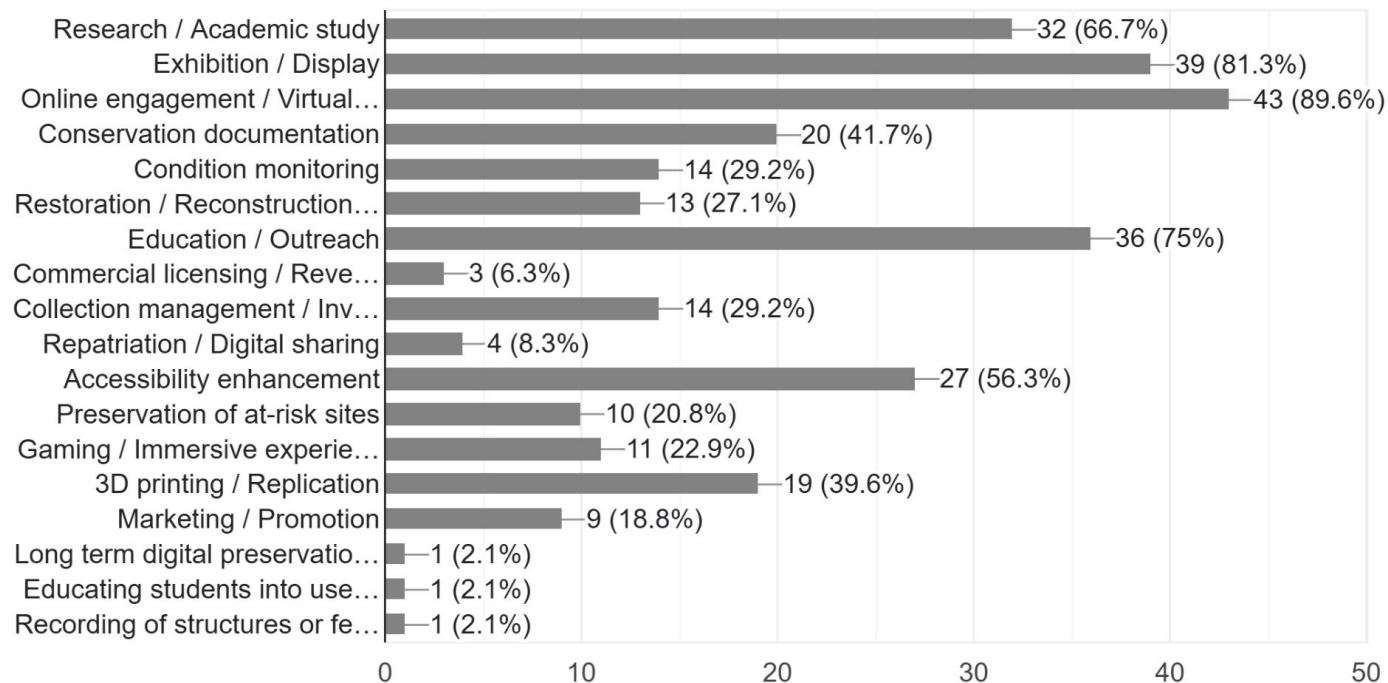
How would you describe the size of your organisation?

48 responses



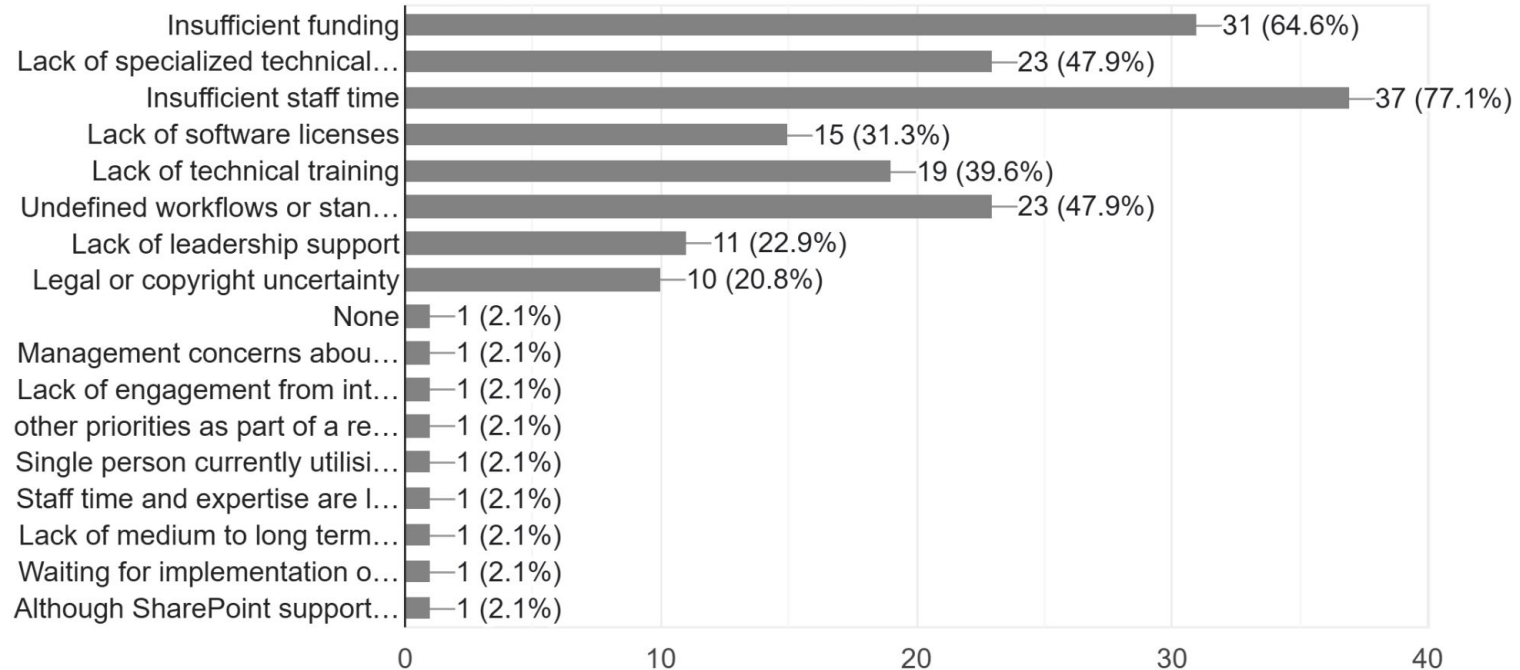
What are your organization's primary goals in producing and publishing 3D data? (Select all that apply)

48 responses



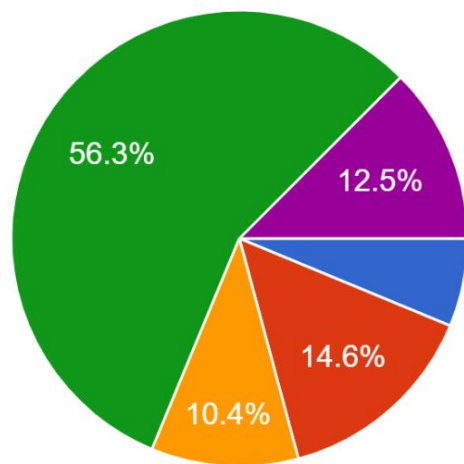
What types of organizational resources—if any—currently limit your ability to engage in or expand high-quality 3D data management? (Select all that apply)

48 responses



Does your organization have a formal Digital Preservation Policy that explicitly covers 3D data types?

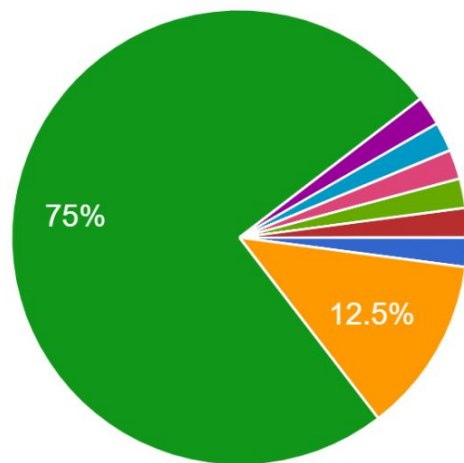
48 responses



- Yes, we have a specific policy for 3D/Complex media.
- Yes, but 3D is only covered under general "digital data" guidelines.
- No, but we are currently drafting one.
- No, we have no formal preservation policy.
- Unknown.

Does your organization have a formal protocol for the 'De-identification' of 3D data prior to publication?

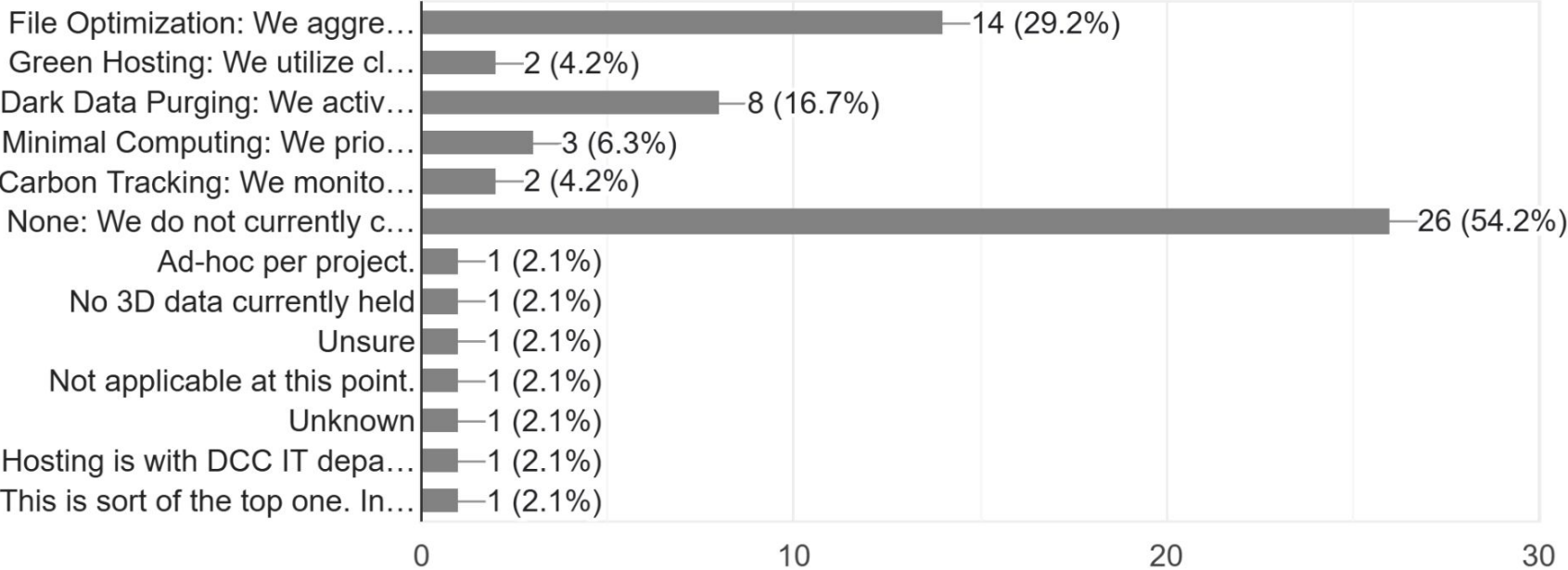
48 responses



- Yes, Automated: We have a software pipeline that automatically scrubs PII.
- Yes, Manual: We have a checklist that...
- Ad-hoc: It is up to the individual resear...
- No: We do not currently have a specifi...
- Unknown
- Sensitive areas of sites will be agreed...
- Unsure
- Not Applicable.
- We do not currently have a specific pr...

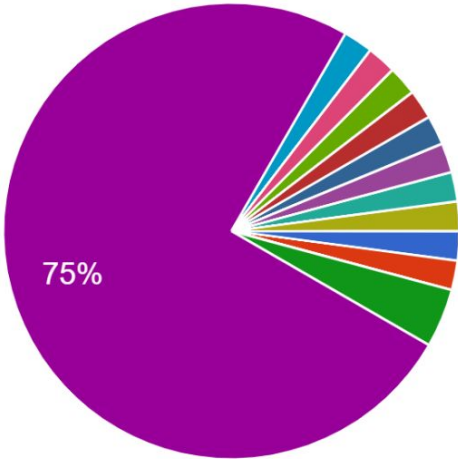
Which of the following sustainable data practices does your organization apply to its 3D workflows? (Select all that apply)

48 responses



How does your organization approach Indigenous Data Sovereignty regarding 3D data?

48 responses



- Source Community Ownership: We re...
- Shared Stewardship: We co-manage t...
- Institutional Ownership: We retain full...
- Open Access Priority: We prioritize glo...
- No Policy: We have not yet addressed...
- N/A
- Beyond the knowledge of my research...
- Ad-hoc per project.

Ethics

Ethics of 3D

“The ethics of archaeology are always dependent on context and historical and social background of the site, work, authors and participants in the project”

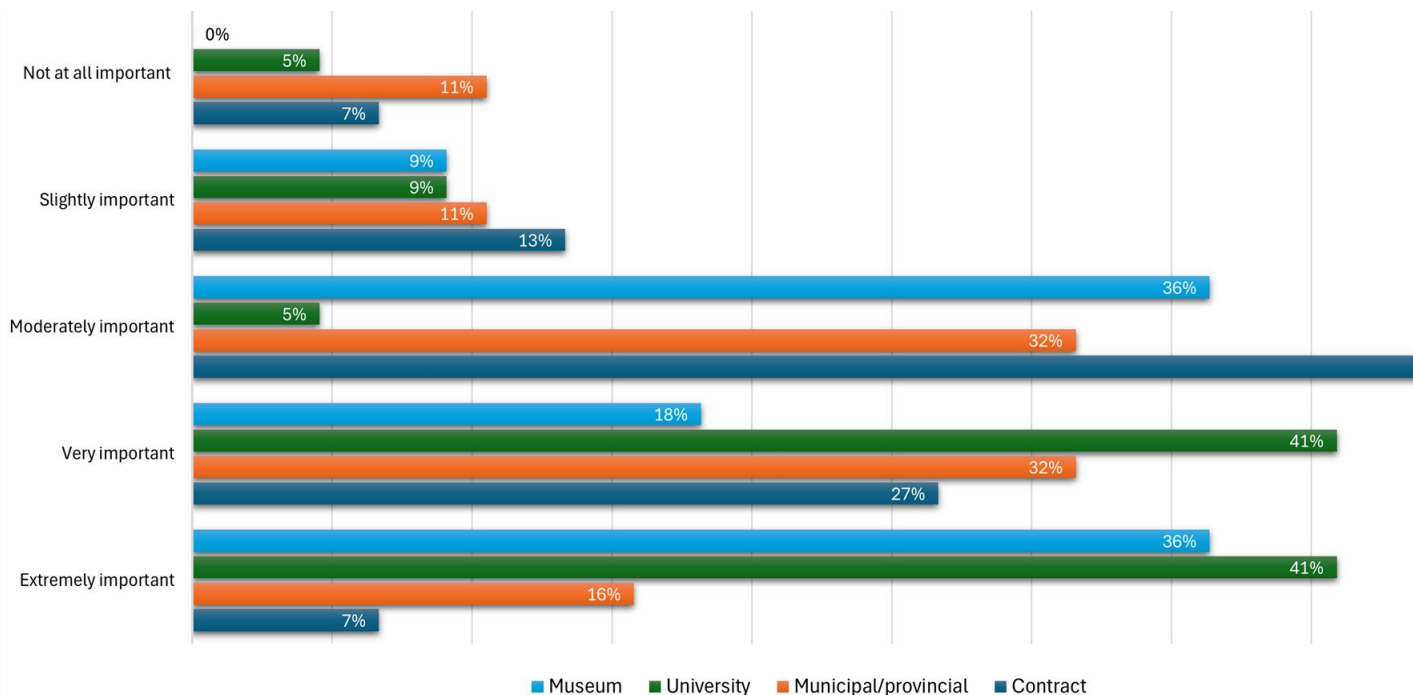
- Richardson, L. 2018



Image: OpenAI, prompt by H. Mickleburgh

Perceived Importance of Digital Ethics

(Archaeologists and heritage professionals in the Netherlands)



Ethics: How to approach and analyze ethics of 3D

Practical: deontological;
professional rules and standards

Philosophical: exploration of
questions about our human
nature

Political: archaeological practice
within systems of power and
society



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Ethics of Archaeology

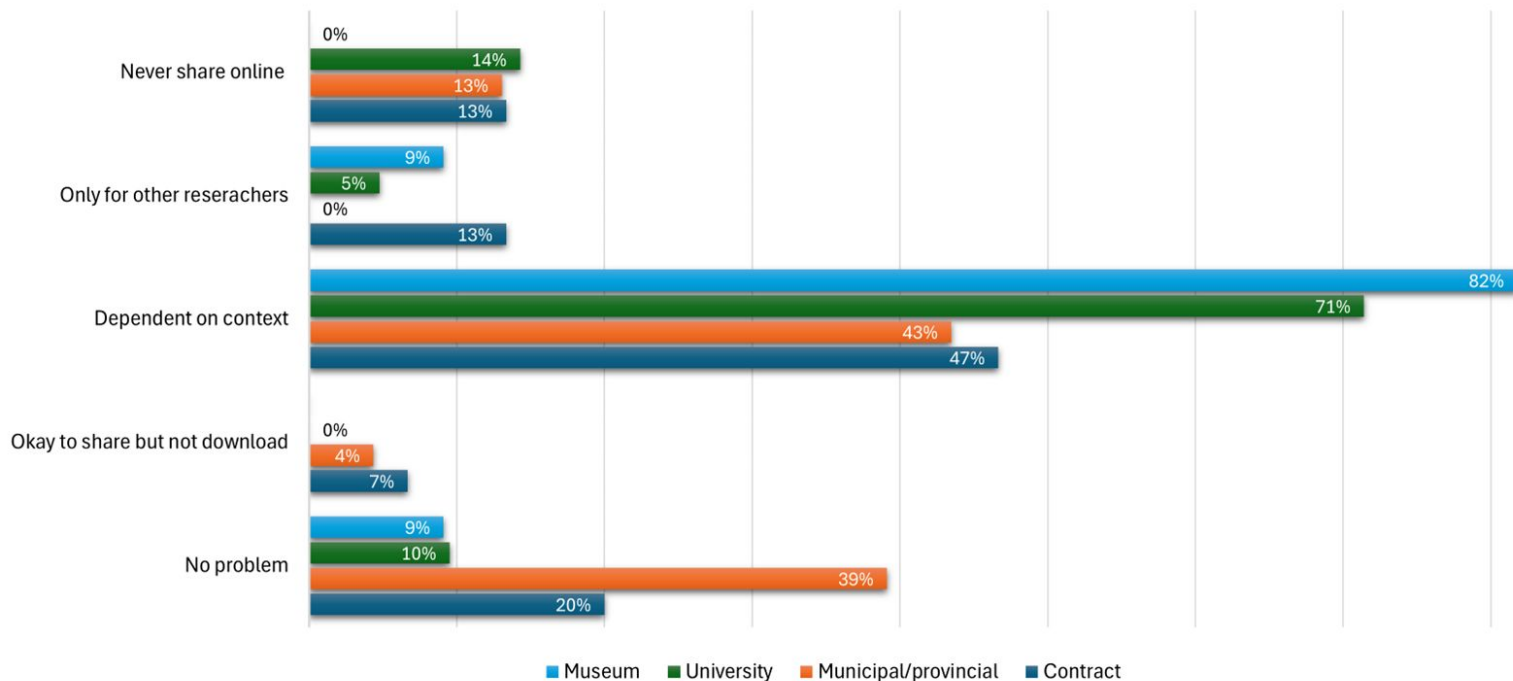
[Alfredo González-Ruibal](#)¹

➕ View Affiliations

Vol. 47:345-360 (Volume publication date October 2018) | <https://doi.org/10.1146/annurev-anthro-102317-045825>

First published as a Review in Advance on August 01, 2018

Ethics of 3D human remains



Bohling, S. (2025) Online dissemination of 3D bioarchaeological data: An exploration of ethics, user preferences, and contextualisation in an official digital repository setting, *Internet Archaeology*

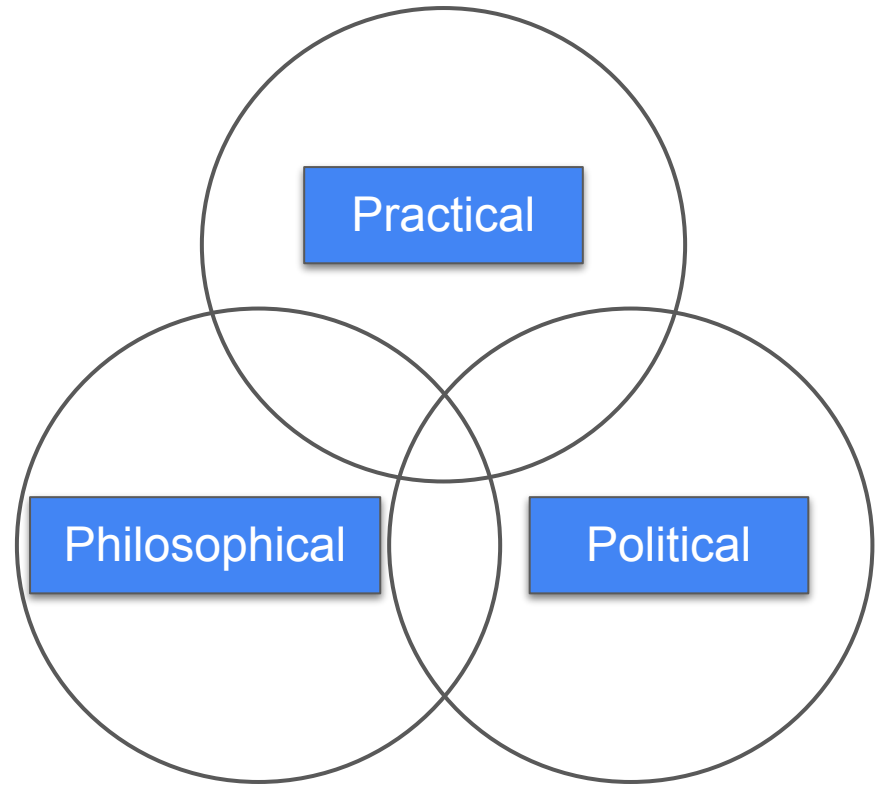
Schug, G.R. et al. (2021) 3D Dead: Ethical Considerations in Digital Human Osteology, *Bioarchaeology International*

Walsh, A. (2026). Preliminary survey data

Identifying the ethics

Interview and survey responses:

- Environmental impact of large data
- Being critical about using these technologies
- Awareness about differing cultural values within a country
- Treatment of human remains
- Codes of conduct
- Data management
- Privacy



tDAR-3D: An Open Science Workshop for Sustainable 3D Cultural Heritage Preservation



Access & Use

Broadening the access to archaeological data through simple search and browse functionality.

[LEARN MORE](#)



Upload Resources

Contribute documents, datasets, images, and other critical archaeological materials.

[LEARN MORE](#)



Preservation

Dedicated to ensuring long-term preservation of digital archaeological data.

[LEARN MORE](#)



Who Uses tDAR

Researchers like you. Uncover knowledge of the past, and preserve and protect resources.

[LEARN MORE](#)



ARCHIVE A FILE FOR AS LOW AS \$10



Featured Content

Isolate record for VAFB-ISO-476

This document is the isolated find record for VAFB-ISO-476. The artifact is a Monterey chert biface margin of gray/brown banded chert.

[VIEW DOCUMENT](#)

or [Browse all Resources](#)



Issues raised:

- A need for clear representations by data producers/collectors addressing CARE principles during data upload/ingestion.
 - “I have permission to upload this data, obtained in this manner...”
 - “There are no known obstructions to publishing this data...”
- A need for *granular* permissions for access, licensing, and labelling of data.
 - Culturally sensitive data, warning messages, authentication, etc.
- Mechanisms for challenges to misuse of content, opening a dialogue.
 - e.g. a [DMCA](#) process, but for ethical challenges.
 - What is the power dynamic? Who decides if a challenge is valid?

Metadata & Paradata

Helping us understand
quality & value.

Engaging, delightful, educational, 3D experiences for audiences.

↑ **powering** ↑

Accessible libraries of openly licensed & interoperable 3D data.

↑ ***underpinning*** ↑

Well **documented** and archived high quality 3D data sets.



Amphore



Ânfora Panatenaica - Panatena...



Amphora



<https://skfb.ly/otZF6>



<https://skfb.ly/p7GDA>



Black-Figure Amp...



Figure Neck Ampho...



el rapto de Casan...



Metadata: "Data About the Data"

Descriptive Metadata (About the CH Object):

- **Title:** e.g., "Venus de Milo"
- **Creator:** e.g., "Alexandros of Antioch"
- **Date:** e.g., "c. 130–100 BC"
- Etc.

Technical/Administrative Metadata (About the 3D Model):

- **File Format:** e.g., ".obj", ".ply", ".gltf"
- **Polygon Count:** e.g., "2.5 million triangles"
- **Texture Resolution:** e.g., "4K (4096x4096px)"
- etc.



Paradata: "Data About the Process"

Capture Data:

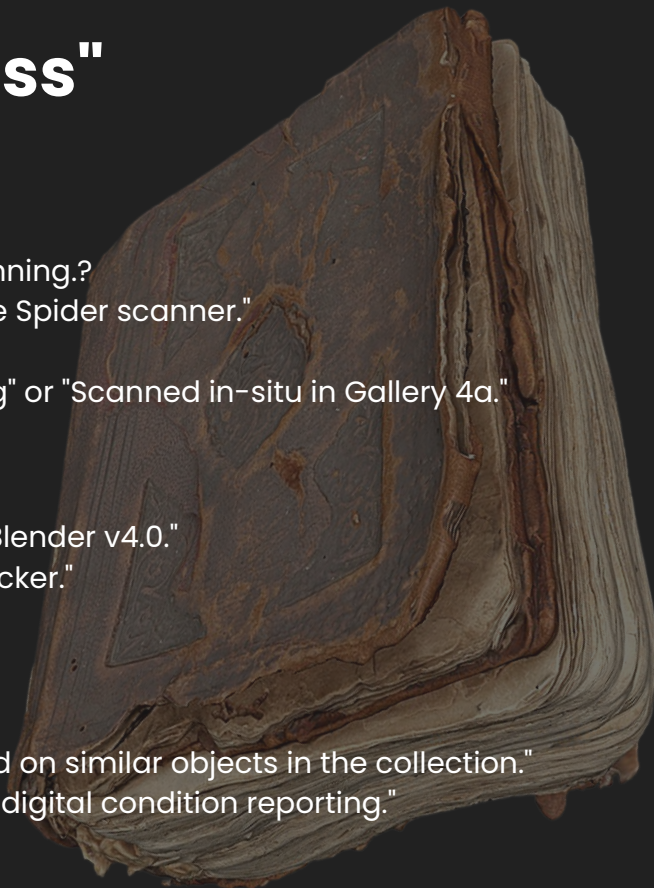
- **Technology Used:** e.g. photogrammetry, structured light, or laser scanning.?
- **Equipment:** e.g., "Sony A7R IV camera with 50mm lens" or "Artec Space Spider scanner."
- **Settings:** e.g., "350 photos taken at f/11, ISO 100."
- **Environment:** e.g., "Captured in a studio with polarized softbox lighting" or "Scanned in-situ in Gallery 4a."

Processing Data:

- **Software Used:** e.g., "Processed in Agisoft Metashape v2.1, cleaned in Blender v4.0."
- **Workflow Steps:** e.g., "Color was normalized using an X-Rite ColorChecker."

Intellectual & Subjective Decisions:

- This is the most critical part of paradata. It answers the "why."
- e.g., "Missing geometry on the back was inferred by the 3D artist based on similar objects in the collection."
- e.g., "Scratches and damage were intentionally preserved to allow for digital condition reporting."



Coptic prayer book by Världskulturmuseerna is dedicated to the public domain

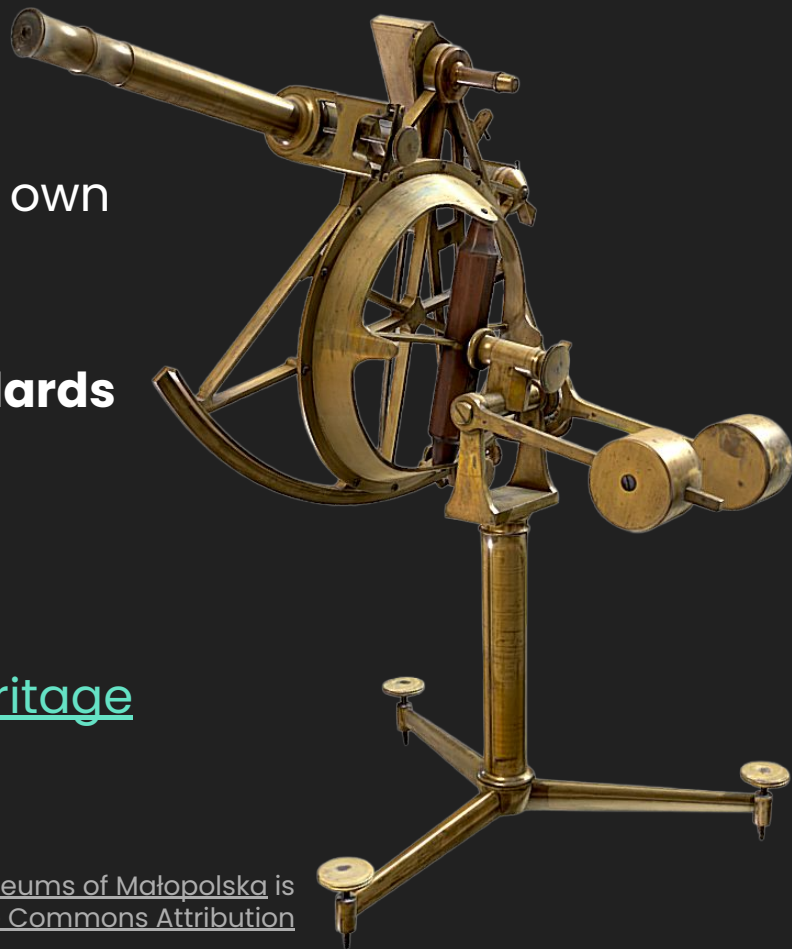
Standards & Guidance

Majority of GLAM organisations use their own unique 3D workflows.

Standards for heritage 3D capture are developing but **only become *true standards* when lots of folks begin to use them.**

- [Community Standards for 3D Data Preservation](#)
- [English Heritage LiDAR](#)
- [UNESCO Chair on Digital Cultural Heritage](#)
- [glam3d.org](#)
- &tc...

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Ideal Paradata in Scholarship

2009:

[The London Charter](#)

2017:

[The Seville Principles](#)

2020:

[glam3d.org](#)

2021-:

[PURE3D: An Infrastructure for the Publication and Preservation of 3D Scholarship](#)

2022:

[3D Data Creation to Curation: Community Standards for 3D Data Preservation](#)

2025:

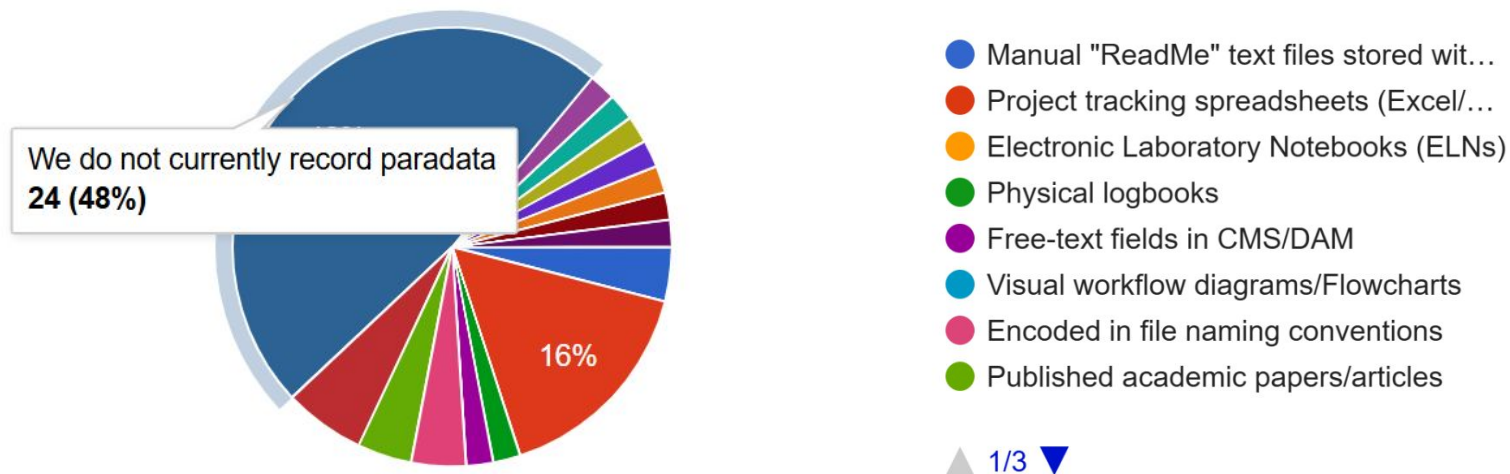
[3D Research Challenges in Cultural Heritage V](#)



Paradata in Practice

How do you capture and link paradata (documentation of the methodological decisions and processes) to the resulting 3D data to ensure transparency and reproducibility?

48 responses



Opportunity: Easy to implement workflows.

- 'Bare minimum' and 'Good, Better, Best' guidance.
- Step by step guides.
- Tools / services / platforms
 - [Digital Lab Notebook](#) by [Cultural Heritage Imaging](#)
- AI?

Practical Exercise

Practical Exercise

Look at examples of models from more than one of these 3D Repositories:

- [Sketchfab](#)
- [Pure3D](#)
- [Open Heritage 3D](#)
- [Global Digital Heritage](#)
- [Iconem](#)

Compare similar 3D model publication pages from these online platforms and review available contextual information (metadata & paradata). **Is it clear:**

- How the 3D data was captured/collected/created?
- What legal and ethical considerations are relevant to the subject?
- How long this resource will be available?
- How well optimised is the 3D asset? (download speed/file size)

What do you think would be a ‘best case scenario’ for documentation of:

1. An archival 3D data package?
2. a 3D publication page?

References

- Bohling, S. (2025). Online dissemination of 3D bioarchaeological data: An exploration of ethics, user preferences, and contextualisation in an official digital repository setting, *Internet Archaeology* 69. <https://doi.org/10.11141/ia.69.9>
- González-Ruibal, A. (2018). Ethics of Archaeology. *Annual Review of Anthropology*, 47, 345–360. <https://www.jstor.org/stable/48550911>
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- Schug, G. R., Killgrove, K., Atkin, A., & Baron, K. (2021). 3D Dead: Ethical Considerations in Digital Human Osteology. *Bioarchaeology International*, 4(3–4), 217–230. <https://doi.org/10.5744/bi.2020.3008>